MATH 243 — Probability and Statistics

Conditional Probability, Total Probability and Bayes' Theorem

Lesson notes · Exam approach · TF practice with solutions · Quick notes (KKS Sir's portion — Probability & ANOVA)

How to Read These Notes

EXAM FOCUS	Exam importance, where it appears in the TF collection
MEMORIZE	Rules, tables and formulas to memorize
TRAP	Common mistakes and exam traps
EXAM-READY ANSWER	Model wording you can write in the exam
NOTE	Extra explanation / clarification
Grey box, monospace	Code, worked tables, derivations and hand-drawn sketches
Centered grey box	Formula to remember / final numerical result
Purple heading / ✓ green	Practice questions / their solutions

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Quick Notes — Conditional Probability and Bayes — revision bullets

NOTE Page numbers refer to Lecture 5.pdf (KKS Sir, 7 pp.), written as Lec 5 p. n. "TF p." = page of MATH 243 All Merged Questions.pdf; "22 batch p." = page of MATH 243 (22 batch).pdf (2023-24).

Exam Map — What the TF Papers Ask

Session (paper)	Questions answered by this note [lesson]
2023-24 (MATH 243) TF p. 22 batch p.1	Q1(b) Multiplication law; box A (5W, 7R) → one ball moved to box B (3W, 8R): P(red from B) (12) [L6] · Q1(c) Party support 45/40/15 %, turnout 65/80/50 %: P(A), P(B), P(C) | did not vote (17) [L6]
2022-23 (MATH 243) TF p. 47	Q1(a) Bag 1 (4W, 3B) → one ball moved unseen to bag 2 (3W, 5B): P(black from bag 2) (8) [L6] · Q1(b) Machines B1, B2, B3 (30/45/25 %, defective 2/3/2 %): P(Bi | defective) (15) [L6]
2019-20 (MATH 245) TF p. 34	Q7(a) Truth serum (90 % / 1 %, 5 % guilty): P(innocent | serum says guilty) (17) [L6]
2018-19 (MATH 245) TF p. 24	Q6(b) Picnic of 20: 5 sunburnt, 8 bitten, 10 safe: P(not bitten | sunburnt), P(sunburnt | bitten) (15) [L5]
2015-16 (MATH 241) TF p. 16	Q8(a) TV show: P(man) 0.25, P(woman) 0.35, P(man | wife) 0.8: P(couple), P(husband | wife does not) (13) [L5]

EXAM FOCUS 7 questions, 97 marks; 5 of them (69 marks) are from 2019-20 onward. A Bayes question (three sources → P(source | event)) appeared in 2019-20, 2022-23 and 2023-24, and a "ball transferred from one bag to another" question in both MATH 243 papers. This is the most reliable 20–30 marks of Section A: learn the table method of Lesson 6 until it is automatic.

Lesson 5 — Conditional Probability and the Multiplication Law

Lec 5 pp. 1–4

1. Definition (Lec 5 p. 1)

The probability of A when it is known that B has occurred is the **conditional probability** P(A | B), read "probability of A given B". In general P(A | B) ≠ P(A).

$$P(A \mid B) = P(A \cap B) / P(B), \quad P(B) \neq 0$$
$$P(B \mid A) = P(A \cap B) / P(A), \quad P(A) \neq 0$$

2. Multiplication Law (Lec 5 pp. 1–2)

$$P(A \cap B) = P(A)\, P(B \mid A) = P(B)\, P(A \mid B)$$
$$P(A_1 \cap A_2 \cap A_3) = P(A_1)\, P(A_2 \mid A_1)\, P(A_3 \mid A_1 \cap A_2)$$

EXAM-READY ANSWER Multiplication law of probability: for two events A and B, the probability of their simultaneous occurrence equals the unconditional probability of one of them multiplied by the conditional probability of the other given that the first has occurred: $P(A \cap B) = P(A)P(B \mid A)$. For k events, $P(A_1 \cap \dots \cap A_k) = P(A_1)P(A_2 \mid A_1) \dots P(A_k \mid A_1 \cap \dots \cap A_{k-1})$. If A and B are independent it reduces to $P(A \cap B) = P(A)P(B)$. (Asked as the theory part of 2023-24 Q1(b).)

3. Worked Examples (Lec 5 pp. 2–4)

Problem 1 — two dice: sum ≥ 10 given a 5 on the first die

A = {(4,6), (5,5), (5,6), (6,4), (6,5), (6,6)}, B = {(5,1), …, (5,6)}, A ∩ B = {(5,5), (5,6)}. P(A | B) = (2/36)/(6/36) = 1/3.

Problem 2 — TV show: P(H) = 0.4, P(W) = 0.5, P(H | W) = 0.7

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(a) P(couple) = P(W ∩ H) = P(W) P(H | W) = 0.5 × 0.7 = 0.35
(b) P(W | H)  = P(W ∩ H) / P(H) = 0.35 / 0.40 = 0.875
(c) P(W ∪ H)  = P(W) + P(H) − P(W ∩ H) = 0.5 + 0.4 − 0.35 = 0.55
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NOTE The slide writes "0.40 + 0.50 − 0.35" in (c); the order does not matter, 0.55 is right.

Problem 3 — toss until a head or three tosses

S = {H, TH, TTH, TTT} with probabilities 1/2, 1/4, 1/8, 1/8. A = tossed three times = {TTH, TTT}; B = no head on the first toss = {TH, TTH, TTT}; P(A ∩ B) = 1/4, P(B) = 1/2, so P(A | B) = 1/2.

Problem 4 — disabled: equal males and females, 5 % of males and 2 % of females disabled

P(M ∩ D) = 0.5 × 0.05 = 0.025, so P(D | M) = 0.025/0.5 = 0.05 — the conditional probability is just the given rate.

TRAP P(A | B) and P(B | A) are different numbers (0.7 vs 0.875 above). Always write which event is the condition before you divide, and divide by the probability of the condition.

Exam Approach

1. Name the events with letters and write down every given number in symbols (P(H | W) = 0.8 …).
2. Find the joint probability first: P(A ∩ B) = P(B)P(A | B).
3. Any other conditional = joint ÷ probability of the new condition; complements via P(A ∩ B^c) = P(A) − P(A ∩ B).
4. Check that every probability lies in [0, 1] and that P(A ∩ B) ≤ min(P(A), P(B)).

Practice — Lesson 5

(a) Actual — 2015-16 Q8(a) (TF p. 16). (b) Actual — 2018-19 Q6(b) (TF p. 24).

(a) The probability that a married man watches a certain television show is 0.25 and that a married woman watches it is 0.35. The probability that a man watches the show, given that his wife does, is 0.8. Find the probability that (i) a married couple watches the show, (ii) a husband watches the show given that his wife does not. (13)

(b) A group of 20 students went on a picnic. 5 got sunburnt, 8 got bitten by mosquitoes and 10 returned home safely without any mishap. What is the probability that (i) a boy who got sunburnt was ignored by the mosquitoes, (ii) a mosquito-bitten boy also got sunburnt? (15)

✓ Solution

(a) Let H = husband watches, W = wife watches: P(H) = 0.25, P(W) = 0.35, P(H | W) = 0.8.

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(i)  P(H ∩ W) = P(W) P(H | W) = 0.35 × 0.8 = 0.28
(ii) P(H | W^c) = P(H ∩ W^c) / P(W^c) = [P(H) − P(H ∩ W)] / [1 − P(W)]
                = (0.25 − 0.28) / 0.65 = −0.0462   ← negative!
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TRAP As printed, the data are inconsistent: P(H ∩ W) = 0.28 is larger than P(H) = 0.25, which is impossible, so (ii) comes out negative. Write (i) = 0.28, then state that the given numbers cannot all be true. The question is the textbook problem with P(H) = 0.4, P(W) = 0.5, P(H | W) = 0.7 (Lec 5 p. 2); with those numbers (ii) = (0.4 − 0.35)/0.5 = 0.10. If P(H) were, e.g., 0.35, (ii) would be (0.35 − 0.28)/0.65 = 0.1077.

Ask the invigilator, or answer with a clearly stated corrected figure — examiners give credit for spotting it.

(b) Let S = sunburnt, M = bitten. Mishap = S ∪ M, so n(S ∪ M) = 20 − 10 = 10.

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n(S ∩ M) = n(S) + n(M) − n(S ∪ M) = 5 + 8 − 10 = 3
              M      M^c    Total
S             3       2       5
S^c           5      10      15
Total         8      12      20
(i)  P(M^c | S) = n(S ∩ M^c)/n(S) = 2/5 = 0.4
(ii) P(S | M)   = n(S ∩ M)/n(M)   = 3/8 = 0.375
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Lesson 6 — Total Probability and Bayes' Theorem

Lec 5 pp. 4–7

1. Theorem of Total Probability (Lec 5 p. 4)

If $A_1, \ldots, A_k$ are mutually exclusive and exhaustive (a partition of S), then for any event D, $D = (D \cap A_1) \cup \ldots \cup (D \cap A_k)$ with disjoint pieces, so

$$\mathbf{P(D) = \Sigma\, P(A_i)\, P(D \mid A_i)}$$

2. Bayes' Theorem and its Proof (Lec 5 pp. 4–5)

$$\mathbf{P(A \mid B) = P(B \mid A)P(A) / [P(B \mid A)P(A) + P(B \mid A^c)P(A^c)]}$$
$$\mathbf{P(A_i \mid D) = P(A_i)P(D \mid A_i) / \Sigma_j\, P(A_j)P(D \mid A_j)}$$

1. From the definition, $P(B \mid A) = P(A \cap B)/P(A)$, so $P(A \cap B) = P(B \mid A)P(A)$ …(1).
2. $B = (B \cap A) \cup (B \cap A^c)$ with disjoint parts, so $P(B) = P(B \mid A)P(A) + P(B \mid A^c)P(A^c)$ …(2) (total probability).
3. $P(A \mid B) = P(A \cap B)/P(B)$; substitute (1) in the numerator and (2) in the denominator. ■

TRAP The slide's middle step (p. 5) writes $P(B \mid A)P(A)/P(A)$ by a slip; the denominator there must be P(B). Write the proof in the three steps above.

3. The Table Method for Every Bayes Question

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Source A_i | prior P(A_i) | likelihood P(D | A_i) | joint P(A_i)P(D | A_i) | posterior = joint / P(D)
-----------------------------------------------------------------------------------------------
   ...     |    ...       |          ...           |          ...           |        ...
Total      |     1        |                        |     P(D) = Σ joint     |         1
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MEMORIZE Prior × likelihood = joint; Σ joint = P(D) (total probability); posterior = joint / P(D). The posteriors must add to 1 — use that as a check.

4. Worked Examples (Lec 5 pp. 5–6)

Problem 1 — blood test: 90 % detection, 5 % false positive, 1 % prevalence

$$\mathbf{P(D \mid +) = (0.01 \times 0.90) / (0.01 \times 0.90 + 0.99 \times 0.05) = 0.009 / 0.0585 = 0.1538}$$

Only about 15 % of positives are diseased: a rare disease gives many false positives.

Problem 2 — coin chooses the container: I has 2 W 7 B, II has 5 W 6 B

$$\mathbf{P(H \mid W) = (½)(2/9) / [(½)(2/9) + (½)(5/11)] = 22/67}$$

Practice problems at the end of Lecture 5 (pp. 6–7) — answers

Problem	P(event)	Posteriors
Problem 3: car 30 %, walk 30 %, bus 40 %; late 3 %, 10 %, 7 %	0.067	car: 0.1343, walk: 0.4478, bus: 0.4179

Problem	P(event)	Posteriors
Problem 3 (centres): A 50 %, B 20 %, C 30 %; positive 2 %, 12 %, 6 %	0.052	A: 0.1923, B: 0.4615, C: 0.3462
Problems 4 and 5: 40 %, 35 %, 25 %; 80 %, 70 %, 90 %	0.79	1st: 0.4051, 2nd: 0.3101, 3rd: 0.2848

5. Transferring a Ball — Total Probability (textbook, asked 2022-23 and 2023-24)

A ball moved unseen from bag 1 to bag 2 creates two cases (the moved ball is W or B). Condition on the case, then use total probability for the second draw.

Exam Approach

1. Define the partition (machines, parties, colour of the moved ball, guilty/innocent) and the observed event D.
2. Fill the prior × likelihood table; add the joints to get P(D).
3. Posterior of each source = its joint ÷ P(D); check that the posteriors sum to 1.
4. Read each percentage carefully and take complements where the event is the opposite one ("did not vote" uses 1 − turnout).

Practice — Lesson 6

(a) Actual — 2022-23 Q1(a) (TF p. 47). (b) Actual — 2023-24 Q1(b) (22 batch p. 1). (c) Actual — 2022-23 Q1(b) (TF p. 47). (d) Actual — 2023-24 Q1(c) (22 batch p. 1). (e) Actual — 2019-20 Q7(a) (TF p. 34).

(a) One bag contains 4 white and 3 black balls, a second bag 3 white and 5 black. One ball is drawn from the first bag and placed unseen in the second. What is the probability that a ball now drawn from the second bag is black? (8)

(b) Define the multiplication law of probability. Box A has 5 white and 7 red balls, box B has 3 white and 8 red. One ball is transferred from A to B unseen, then a ball is drawn from B. What is the probability that it is red? (12)

(c) Machines B_1, B_2, B_3 make 30 %, 45 % and 25 % of the products; 2 %, 3 % and 2 % of their products are defective. A finished product is selected at random and is defective. What is the probability that it was made by machine B_i, i = 1, 2, 3? (15)

(d) 45 % of a city support party A, 40 % party B and 15 % other parties C. In elections 65 % of A, 80 % of B and 50 % of C supporters vote. A person chosen at random did not vote in the last election. What is the probability that he is a supporter of A? B? C? (17)

(e) A truth serum judges 90 % of guilty suspects properly (10 % of the guilty are found innocent); innocent suspects are misjudged 1 % of the time. If the suspect comes from a group of which only 5 % have committed a crime and the serum indicates guilty, what is the probability that he is innocent? (17)

✓ Solution

(a) Let W_1, B_1 = colour of the moved ball; Bk = black drawn from bag 2.

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Case W1: P(W1) = 4/7, bag 2 becomes 4W 5B → P(Bk | W1) = 5/9
Case B1: P(B1) = 3/7, bag 2 becomes 3W 6B → P(Bk | B1) = 6/9
P(Bk) = (4/7)(5/9) + (3/7)(6/9) = 20/63 + 18/63 = 38/63 = 0.6032
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(b) Multiplication law as in Lesson 5 §2. Then condition on the colour of the transferred ball:

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Case W: P(W) = 5/12, box B becomes 4W 8R → P(R | W) = 8/12
Case R: P(R) = 7/12, box B becomes 3W 9R → P(R | R) = 9/12
P(red from B) = (5/12)(8/12) + (7/12)(9/12) = 40/144 + 63/144 = 103/144 = 0.7153
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(c) D = defective.

Machine	Prior $P(B_i)$	$P(D \mid B_i)$	Joint	$P(B_i \mid D)$
B1	0.3	0.02	0.006	0.006/0.0245 = 0.2449
B2	0.45	0.03	0.0135	0.0135/0.0245 = 0.551
B3	0.25	0.02	0.005	0.005/0.0245 = 0.2041
Total	1		P(D) = 0.0245	1

P(B1 | D) = 0.2449, P(B2 | D) = 0.551, P(B3 | D) = 0.2041 (= 12/49, 27/49, 10/49)

(d) N = did not vote; P(N | A) = 1 − 0.65 = 0.35, P(N | B) = 0.20, P(N | C) = 0.50.

Party	Prior	P(N | party)	Joint	P(party | N)
A	0.45	0.35	0.1575	0.1575/0.3125 = 0.504
B	0.4	0.2	0.08	0.08/0.3125 = 0.256
C	0.15	0.5	0.075	0.075/0.3125 = 0.24
Total	1		P(N) = 0.3125	1

P(A | N) = 0.504, P(B | N) = 0.256, P(C | N) = 0.24

(e) G = guilty (prior 0.05), I = innocent (0.95); J = serum says guilty: P(J | G) = 0.90, P(J | I) = 0.01.

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P(J) = 0.05 × 0.90 + 0.95 × 0.01 = 0.045 + 0.0095 = 0.0545
P(I | J) = 0.0095 / 0.0545 = 0.1743
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P(innocent | judged guilty) ≈ 0.174

Quick Notes — Conditional Probability and Bayes

Revision points for Lessons 5–6.

L5 · Conditional Probability

- $P(A \mid B) = P(A \cap B)/P(B)$. Multiplication law: $P(A \cap B) = P(A)P(B \mid A)$.
- Three events: $P(A_1)P(A_2 \mid A_1)P(A_3 \mid A_1 \cap A_2)$.
- TV couple (0.4, 0.5, 0.7): both 0.35, $P(W \mid H) = 0.875$, at least one 0.55.
- Check $P(A \cap B) \le \min(P(A), P(B))$ — 2015-16 Q8(a) data fail this test.
- Picnic: both = 5 + 8 − 10 = 3 → 2/5 and 3/8.

L6 · Total Probability and Bayes

- $P(D) = \Sigma P(A_i)P(D \mid A_i)$; $P(A_i \mid D)$ = joint/$P(D)$.
- Proof: definition → total probability → substitute.
- Blood test 0.15; machines 12/49, 27/49, 10/49; parties 0.504, 0.256, 0.24 ($P(N) = 0.3125$).
- Truth serum: $P(J) = 0.0545$, $P(\text{innocent} \mid J) = 0.1743$.
- Transfer: 4W3B → 3W5B gives 38/63; 5W7R → 3W8R gives 103/144.